# Лабораторная работа №8

Адресация IPv4 и IPv6. Настройка маршрутизации

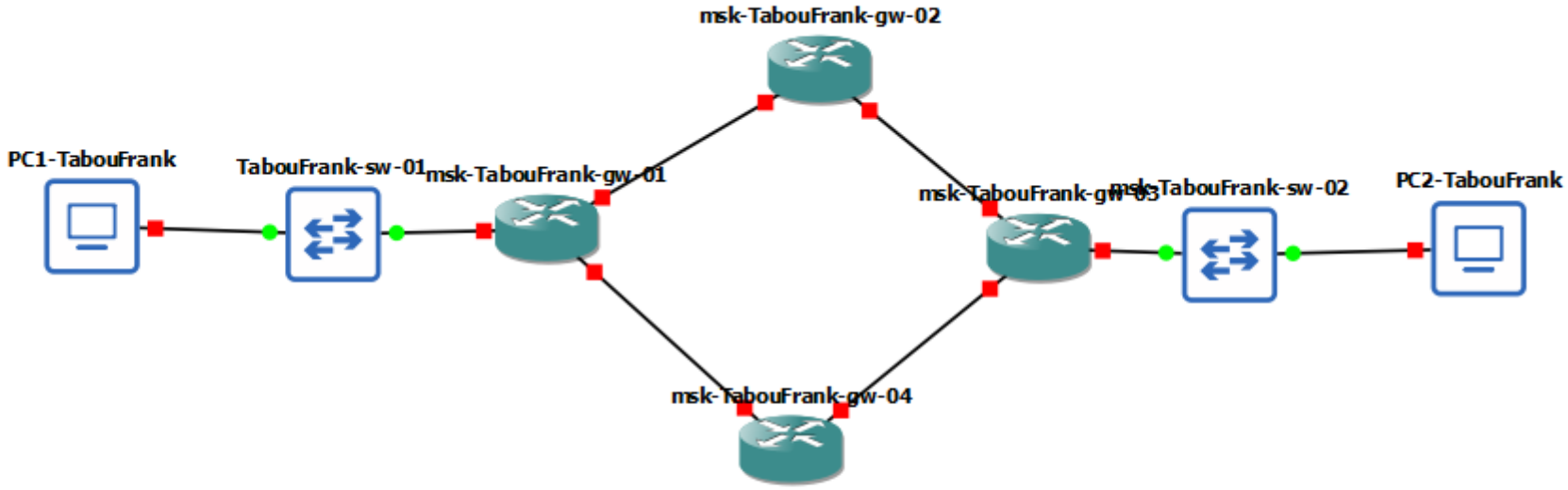
**Студент: Талебу Тенке Франк Устон**

**Группа: НФИбд 02–23**

***дисциплина:*** **Сетевые технологии**

# Цель работы

Изучение принципов маршрутизации в IPv4- и IPv6-сетях и принципов настройки сетевого оборудования.


msk-TabouFrank-gw-02
PC1-TabouFrank
TabouFrank-sw-01
msk-TabouFrank-gw-01
msk-TabouFrank-gw-03
msk-TabouFrank-sw-02
PC2-TabouFrank
msk-TabouFrank-gw-04

PC2-TabouFrank - PuTTY

```
VPCS> ip 10.0.11.10/24 10.0.11.1
Checking for duplicate address...
VPCS : 10.0.11.10 255.255.255.0 gateway 10.0.11.1

VPCS> save
Saving startup configuration to startup.vpc
.  done

VPCS> shop ip
Bad command: "shop ip". Use ? for help.

VPCS> show ip

NAME        : VPCS[1]
IP/MASK     : 10.0.11.10/24
GATEWAY     : 10.0.11.1
DNS         :
MAC         : 00:50:79:66:68:01
LPORT       : 10036
RHOST:PORT  : 127.0.0.1:10037
MTU         : 1500

VPCS>
```

```
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-frank-gw-01
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.10.1/24
exit
!
interface eth1
 ip address 10.0.1.1/24
exit
!
interface eth2
 ip address 10.0.4.2/24
exit
!
end
msk-frank-gw-01#
```

msk-frank-gw-04 - PuTTY

```
Hello, this is FRRouting (version 8.2.2).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

frr# configure terminal
frr(config)# hostname-frank-gw-04
% Unknown command: hostname-frank-gw-04
frr(config)# hostname msk-frank-gw-04
msk-frank-gw-04(config)# interface eth0
msk-frank-gw-04(config-if)# ip address 10.0.3.2/24
msk-frank-gw-04(config-if)# no shutdown
msk-frank-gw-04(config-if)# exit
msk-frank-gw-04(config)# exit
msk-frank-gw-04# configure terminal
msk-frank-gw-04(config)# interface eth1
msk-frank-gw-04(config-if)# ip address 10.0.4.1/24
msk-frank-gw-04(config-if)# no shutdown
msk-frank-gw-04(config-if)# exit
msk-frank-gw-04(config)# exit
msk-frank-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-04# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-frank-gw-04
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.3.2/24
exit
!
interface eth1
 ip address 10.0.4.1/24
exit
!
end
msk-frank-gw-04#
```

PC2-TabouFrank - PuTTY

```
LPORT        : 10036
RHOST:PORT   : 127.0.0.1:10037
MTU          : 1500

VPCS> ip 2001:11::a/64
PC1 : 2001:11::a/64

VPCS> save
Saving startup configuration to startup.vpc
.  done

VPCS> show ipv6

NAME                : VPCS[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE        : 2001:11::a/64
DNS                 :
ROUTER LINK-LAYER   :
MAC                 : 00:50:79:66:68:01
LPORT               : 10036
RHOST:PORT          : 127.0.0.1:10037
MTU:                : 1500

VPCS>
```

msk-frank-gw-02 - PuTTY

```
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.2.1/24
exit
!
end
msk-frank-gw-02# configure terminal
msk-frank-gw-02(config)# ipv6 forwarding
msk-frank-gw-02(config)# interface eth0
msk-frank-gw-02(config-if)# ipv6 address 2001:1::2/64
msk-frank-gw-02(config-if)# no shutdown
msk-frank-gw-02(config-if)# exit
msk-frank-gw-02(config)# interface eth1
msk-frank-gw-02(config-if)# ipv6 address 2001:2::1/64
msk-frank-gw-02(config-if)# no shutdown
msk-frank-gw-02(config-if)# exit
msk-frank-gw-02(config)# exit
msk-frank-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-02#
```

msk-frank-gw-03 - PuTTY

```
service integrated-vtysh-config
!
interface eth1
 ip address 10.0.2.2/24
exit
!
interface eth2
 ip address 10.0.3.1/24
exit
!
interface eth0
 ip address 10.0.11.1/24
exit
!
end
msk-frank-gw-03# configure interface
% Unknown command: configure interface
msk-frank-gw-03# configure terminal
msk-frank-gw-03(config)# ipv6 forwarding
msk-frank-gw-03(config)# interface eth0
msk-frank-gw-03(config-if)# ipv6 address 2001:11::1/64
msk-frank-gw-03(config-if)# no ivp6 nd suppress-ra
% Unknown command: no ivp6 nd suppress-ra
msk-frank-gw-03(config-if)# no ipv6 nd suppress-ra
msk-frank-gw-03(config-if)# ipv6 nd prefix 2001:11::/64
msk-frank-gw-03(config-if)# no shutdown
msk-frank-gw-03(config-if)# exit
msk-frank-gw-03(config)# interface eth1
msk-frank-gw-03(config-if)# ipv6 address 2001:2::2/64
msk-frank-gw-03(config-if)# no shutdown
msk-frank-gw-03(config-if)# exit
msk-frank-gw-03(config)#
```

*Standard input [msk-frank-gw-03 eth0 to msk-TabouFrank-sw-02 Ethernet1]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Примените фильтр отображения ... <Ctrl-/>

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 8 | 4203.576917 | fe80::e50:bdff:fe60… | ff02::1 | ICMPv6 | 110 | Router Advertisement from 0c:50:bd:60:00:00 |
| 9 | 4804.241934 | fe80::e50:bdff:fe60… | ff02::1 | ICMPv6 | 110 | Router Advertisement from 0c:50:bd:60:00:00 |
| 10 | 5405.085094 | fe80::e50:bdff:fe60… | ff02::1 | ICMPv6 | 110 | Router Advertisement from 0c:50:bd:60:00:00 |
| 11 | 6005.851192 | fe80::e50:bdff:fe60… | ff02::1 | ICMPv6 | 110 | Router Advertisement from 0c:50:bd:60:00:00 |
| 12 | 6606.339404 | fe80::e50:bdff:fe60… | ff02::1 | ICMPv6 | 110 | Router Advertisement from 0c:50:bd:60:00:00 |
| 13 | 7208.098765 | fe80::e50:bdff:fe60… | ff02::1 | ICMPv6 | 110 | Router Advertisement from 0c:50:bd:60:00:00 |
| 14 | 7807.759946 | fe80::e50:bdff:fe60… | ff02::1 | ICMPv6 | 110 | Router Advertisement from 0c:50:bd:60:00:00 |
| 15 | 8408.660375 | fe80::e50:bdff:fe60… | ff02::1 | ICMPv6 | 110 | Router Advertisement from 0c:50:bd:60:00:00 |
| 16 | 9009.094382 | fe80::e50:bdff:fe60… | ff02::1 | ICMPv6 | 110 | Router Advertisement from 0c:50:bd:60:00:00 |
| 17 | 9609.884208 | fe80::e50:bdff:fe60… | ff02::1 | ICMPv6 | 110 | Router Advertisement from 0c:50:bd:60:00:00 |
| 18 | 10213.137096 | fe80::e50:bdff:fe60… | ff02::1 | ICMPv6 | 110 | Router Advertisement from 0c:50:bd:60:00:00 |
| 19 | 10812.537844 | fe80::e50:bdff:fe60… | ff02::1 | ICMPv6 | 110 | Router Advertisement from 0c:50:bd:60:00:00 |
| 20 | 11412.960835 | fe80::e50:bdff:fe60… | ff02::1 | ICMPv6 | 110 | Router Advertisement from 0c:50:bd:60:00:00 |
| 21 | 12013.266510 | fe80::e50:bdff:fe60… | ff02::1 | ICMPv6 | 110 | Router Advertisement from 0c:50:bd:60:00:00 |
| 22 | 12613.799163 | fe80::e50:bdff:fe60… | ff02::1 | ICMPv6 | 110 | Router Advertisement from 0c:50:bd:60:00:00 |
| 23 | 13214.308867 | fe80::e50:bdff:fe60… | ff02::1 | ICMPv6 | 110 | Router Advertisement from 0c:50:bd:60:00:00 |
| 24 | 13814.715193 | fe80::e50:bdff:fe60… | ff02::1 | ICMPv6 | 110 | Router Advertisement from 0c:50:bd:60:00:00 |
| 25 | 14415.102113 | fe80::e50:bdff:fe60… | ff02::1 | ICMPv6 | 110 | Router Advertisement from 0c:50:bd:60:00:00 |
| 26 | 15015.717395 | fe80::e50:bdff:fe60… | ff02::1 | ICMPv6 | 110 | Router Advertisement from 0c:50:bd:60:00:00 |
| 27 | 15616.353197 | fe80::e50:bdff:fe60… | ff02::1 | ICMPv6 | 110 | Router Advertisement from 0c:50:bd:60:00:00 |
| 28 | 16217.002846 | fe80::e50:bdff:fe60… | ff02::1 | ICMPv6 | 110 | Router Advertisement from 0c:50:bd:60:00:00 |
| 29 | 16817.714618 | fe80::e50:bdff:fe60… | ff02::1 | ICMPv6 | 110 | Router Advertisement from 0c:50:bd:60:00:00 |
| 30 | 17418.478152 | fe80::e50:bdff:fe60… | ff02::1 | ICMPv6 | 110 | Router Advertisement from 0c:50:bd:60:00:00 |
| 31 | 18019.082684 | fe80::e50:bdff:fe60… | ff02::1 | ICMPv6 | 110 | Router Advertisement from 0c:50:bd:60:00:00 |
| 32 | 18619.710044 | fe80::e50:bdff:fe60… | ff02::1 | ICMPv6 | 110 | Router Advertisement from 0c:50:bd:60:00:00 |
| 33 | 19220.346729 | fe80::e50:bdff:fe60… | ff02::1 | ICMPv6 | 110 | Router Advertisement from 0c:50:bd:60:00:00 |
| 34 | 19821.088089 | fe80::e50:bdff:fe60… | ff02::1 | ICMPv6 | 110 | Router Advertisement from 0c:50:bd:60:00:00 |

```
> Frame 1: 110 bytes on wire (880 bits), 110 bytes captured (880 bits) on interface -, id 0
> Ethernet II, Src: 0c:50:bd:60:00:00 (0c:50:bd:60:00:00), Dst: IPv6mcast_01 (33:33:00:00:00:01)
> Internet Protocol Version 6, Src: fe80::e50:bdff:fe60:0, Dst: ff02::1
> Internet Control Message Protocol v6
```

```
0000  33 33 00 00 00 01 0c 50  bd 60 00 00 86 dd 60 0f   33·····P ····`·
0010  29 37 00 38 3a ff fe 80  00 00 00 00 00 00 0e 50   )7·8:··· ·····P
0020  bd ff fe 60 00 00 ff 02  00 00 00 00 00 00 00 00   ···`···· ····
0030  00 00 00 00 00 01 86 00  74 16 40 00 07 08 00 00   ····· t
0040  00 00 00 00 00 00 03 04  40 c0 00 27 8d 00 00 09   ····· @
0050  3a 80 00 00 00 00 20 01  00 11 00 00 00 00 00 00   :····· ····
0060  00 00 00 00 00 00 01 01  0c 50 bd 60 00 00         ····· ·
```

wireshark_Standard input83C5L3.pcapng | Пакеты: 34 · Потеряно: 0 (0.0%) | Профиль: Default

msk-frank-gw-01 - PuTTY

```
frr(config-router)# network eth1
frr(config-router)# exit
frr(config)# exit
frr# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
frr# configure terminal
frr(config)# hostname msk-frank-gw-01
msk-frank-gw-01(config)# show ipv6 ripng
% Unknown command: show ipv6 ripng
msk-frank-gw-01(config)# router ospf
msk-frank-gw-01(config-router)# network 10.0.10.0/24 area 0.0.0.0
msk-frank-gw-01(config-router)# network 10.0.1.0/24 area 0.0.0.0
msk-frank-gw-01(config-router)# network 10.0.4.0/24 area 0.0.0.0
msk-frank-gw-01(config-router)# exit
msk-frank-gw-01(config)# exit
msk-frank-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-01#
```

Захват из Standard input [TabouFrank-sw-01 Ethernet1 to msk-frank-gw-01 eth0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Примените фильтр отображения ... <Ctrl-/>

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 1 | 0.000000 | 10.0.10.1 | 224.0.0.9 | RIPv2 | 126 | Response |
| 2 | 22.820365 | fe80::e2e:91ff:fe2f… | ff02::9 | RIPng | 106 | Command Response, Version 1 |
| 3 | 34.006925 | 10.0.10.1 | 224.0.0.9 | RIPv2 | 126 | Response |
| 4 | 67.011417 | 10.0.10.1 | 224.0.0.9 | RIPv2 | 126 | Response |
| 5 | 67.824948 | fe80::e2e:91ff:fe2f… | ff02::9 | RIPng | 106 | Command Response, Version 1 |
| 6 | 101.014537 | 10.0.10.1 | 224.0.0.9 | RIPv2 | 126 | Response |
| 7 | 109.826729 | fe80::e2e:91ff:fe2f… | ff02::9 | RIPng | 106 | Command Response, Version 1 |
| 8 | 131.017360 | 10.0.10.1 | 224.0.0.9 | RIPv2 | 126 | Response |
| 9 | 133.570590 | fe80::e2e:91ff:fe2f… | ff02::1 | ICMPv6 | 110 | Router Advertisement from 0c:2e:91:2f:00:00 |
| 10 | 151.830745 | fe80::e2e:91ff:fe2f… | ff02::9 | RIPng | 106 | Command Response, Version 1 |
| 11 | 155.019884 | 10.0.10.1 | 224.0.0.9 | RIPv2 | 126 | Response |
| 12 | 167.836251 | fe80::e2e:91ff:fe2f… | ff02::9 | RIPng | 106 | Command Response, Version 1 |
| 13 | 189.025037 | 10.0.10.1 | 224.0.0.9 | RIPv2 | 126 | Response |
| 14 | 209.837058 | fe80::e2e:91ff:fe2f… | ff02::9 | RIPng | 106 | Command Response, Version 1 |
| 15 | 225.026118 | 10.0.10.1 | 224.0.0.9 | RIPv2 | 126 | Response |
| 16 | 245.844109 | fe80::e2e:91ff:fe2f… | ff02::9 | RIPng | 106 | Command Response, Version 1 |

```
> Frame 1: 126 bytes on wire (1008 bits), 126 bytes captured (1008 bits) on interface -, id 0
> Ethernet II, Src: 0c:2e:91:2f:00:00 (0c:2e:91:2f:00:00), Dst: IPv4mcast_09 (01:00:5e:00:00:09)
> Internet Protocol Version 4, Src: 10.0.10.1, Dst: 224.0.0.9
> User Datagram Protocol, Src Port: 520, Dst Port: 520
> Routing Information Protocol
```

```
0000  01 00 5e 00 00 09 0c 2e  91 2f 00 00 08 00 45 c0   ··^·····  ·
0010  00 70 ba 10 40 00 01 11  ca a2 0a 00 0a 01 e0 00   ·p··@···  ·
0020  00 09 02 08 02 08 00 5c  d7 06 02 02 00 00 00 02   ·******\  ·
0030  00 00 0a 00 01 00 ff ff  ff 00 00 00 00 00 00 00   ********  ·
0040  00 01 00 02 00 00 0a 00  02 00 ff ff ff 00 00 00   ********  ·
0050  00 00 00 00 00 03 00 02  00 00 0a 00 03 00 ff ff   ********  ·
0060  ff 00 00 00 00 00 00 00  00 02 00 02 00 00 0a 00   ********  ·
0070  04 00 ff ff ff 00 00 00  00 00 00 00 00 01         ******··  ·
```

msk-frank-gw-04 - PuTTY

```
frr(config-router)# exit
frr(config)# exit
frr# write memoey
% Unknown command: write memoey
frr# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
frr# cofigure terminal
% Unknown command: cofigure terminal
frr# configure terminal
frr(config)# hostname msk-frank-gw-04
msk-frank-gw-04(config)# router ospf
msk-frank-gw-04(config-router)# network 10.0.3.0/24 area 0.0.0.0
msk-frank-gw-04(config-router)# network 10.0.4.0/24 area 0.0.0.0
msk-frank-gw-04(config-router)# exit
msk-frank-gw-04(config)# exit
msk-frank-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-04#
```

Захват из Standard input [TabouFrank-sw-01 Ethernet1 to msk-frank-gw-01 eth0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Примените фильтр отображения ... <Ctrl-/>

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 137 | 930.682182 | 10.0.10.10 | 10.0.11.10 | TCP | 74 | [TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422786 TSecr=0 WS=2 |
| 138 | 931.682229 | 10.0.10.10 | 10.0.11.10 | TCP | 74 | [TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422787 TSecr=0 WS=2 |
| 139 | 932.682397 | 10.0.10.10 | 10.0.11.10 | TCP | 74 | [TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422788 TSecr=0 WS=2 |
| 140 | 933.560457 | 10.0.10.1 | 224.0.0.5 | OSPF | 78 | Hello Packet |
| 141 | 933.682987 | 10.0.10.10 | 10.0.11.10 | TCP | 74 | [TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422789 TSecr=0 WS=2 |
| 142 | 933.747833 | 0c:2e:91:2f:00:00 | Private_66:68:00 | ARP | 60 | Who has 10.0.10.10? Tell 10.0.10.1 |
| 143 | 933.748224 | Private_66:68:00 | 0c:2e:91:2f:00:00 | ARP | 60 | 10.0.10.10 is at 00:50:79:66:68:00 |
| 144 | 934.684139 | 10.0.10.10 | 10.0.11.10 | TCP | 74 | [TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422790 TSecr=0 WS=2 |
| 145 | 935.684067 | 10.0.10.10 | 10.0.11.10 | TCP | 74 | [TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422791 TSecr=0 WS=2 |
| 146 | 936.684747 | 10.0.10.10 | 10.0.11.10 | TCP | 74 | [TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422792 TSecr=0 WS=2 |
| 147 | 937.685902 | 10.0.10.10 | 10.0.11.10 | TCP | 74 | [TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422793 TSecr=0 WS=2 |
| 148 | 938.686957 | 10.0.10.10 | 10.0.11.10 | TCP | 74 | [TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422794 TSecr=0 WS=2 |
| 149 | 939.687504 | 10.0.10.10 | 10.0.11.10 | TCP | 74 | [TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422795 TSecr=0 WS=2 |
| 150 | 940.687383 | 10.0.10.10 | 10.0.11.10 | TCP | 74 | [TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422796 TSecr=0 WS=2 |
| 151 | 941.687941 | 10.0.10.10 | 10.0.11.10 | TCP | 74 | [TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422797 TSecr=0 WS=2 |
| 152 | 942.688823 | 10.0.10.10 | 10.0.11.10 | TCP | 74 | [TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422798 TSecr=0 WS=2 |
| 153 | 943.560778 | 10.0.10.1 | 224.0.0.5 | OSPF | 78 | Hello Packet |
| 154 | 943.689516 | 10.0.10.10 | 10.0.11.10 | TCP | 74 | [TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422799 TSecr=0 WS=2 |
| 155 | 943.923680 | fe80::e2e:91ff:fe2f… | ff02::9 | RIPng | 106 | Command Response, Version 1 |
| 156 | 944.690720 | 10.0.10.10 | 10.0.11.10 | TCP | 74 | [TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422800 TSecr=0 WS=2 |
| 157 | 945.691550 | 10.0.10.10 | 10.0.11.10 | TCP | 74 | [TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422801 TSecr=0 WS=2 |
| 158 | 946.692577 | 10.0.10.10 | 10.0.11.10 | TCP | 74 | [TCP Retransmission] 57440 → 57441 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1773422802 TSecr=0 WS=2 |

```
Frame 1: 126 bytes on wire (1008 bits), 126 bytes captured (1008 bits) on interface -, id 0
    Section number: 1
    Interface id: 0 (-)
    Encapsulation type: Ethernet (1)
    Arrival Time: Mar 13, 2026 20:10:55.995775000 RTZ 2 (зима)
    UTC Arrival Time: Mar 13, 2026 17:10:55.995775000 UTC
    Epoch Arrival Time: 1773421855.995775000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 0.000000000 seconds]
    [Time delta from previous displayed frame: 0.000000000 seconds]
    [Time since reference or first frame: 0.000000000 seconds]
    Frame Number: 1
    Frame Length: 126 bytes (1008 bits)
    Capture Length: 126 bytes (1008 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
```

```
0000  01 00 5e 00 00 09 0c 2e  91 2f 00 00 08 00 45 c0   ··^·····  ·
0010  00 70 ba 10 40 00 01 11  ca a2 0a 00 0a 01 e0 00   ·p··@···  ·
0020  00 09 02 08 02 08 00 5c  d7 06 02 02 00 00 00 02   ·******\  ·
0030  00 00 0a 00 01 00 ff ff  ff 00 00 00 00 00 00 00   ********  ·
0040  00 01 00 02 00 00 0a 00  02 00 ff ff ff 00 00 00   ********  ·
0050  00 00 00 00 00 03 00 02  00 00 0a 00 03 00 ff ff   ********  ·
0060  ff 00 00 00 00 00 00 00  00 02 00 02 00 00 0a 00   ********  ·
0070  04 00 ff ff ff 00 00 00  00 00 00 00 00 01         ********  ·
```

msk-frank-gw-01 - PuTTY

```
                            directly attached to eth2
N    10.0.10.0/24           [100] area: 0.0.0.0
                            directly attached to eth0

============ OSPF router routing table =============

============ OSPF external routing table ===========


msk-frank-gw-01# show ip ospf neighbor

Neighbor ID     Pri State           Up Time         Dead Time Address         Interface
          RXmtL RqstL DBsmL
10.0.4.1          1 Full/Backup     21m39s            38.604s 10.0.4.1        eth2:10.0.4.2
              0     0     0

msk-frank-gw-01# show ip ospf route
============ OSPF network routing table ============
N    10.0.1.0/24            [100] area: 0.0.0.0
                            directly attached to eth1
N    10.0.3.0/24            [200] area: 0.0.0.0
                            via 10.0.4.1, eth2
N    10.0.4.0/24            [100] area: 0.0.0.0
                            directly attached to eth2
N    10.0.10.0/24           [100] area: 0.0.0.0
                            directly attached to eth0

============ OSPF router routing table =============

============ OSPF external routing table ===========


msk-frank-gw-01#
```

msk-frank-gw-02 - PuTTY

```
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-02# configure terminal
msk-frank-gw-02(config)# interface eht0
msk-frank-gw-02(config-if)# shutdown
msk-frank-gw-02(config-if)# exit
msk-frank-gw-02(config)# interface eht0
msk-frank-gw-02(config-if)# no shutdown
msk-frank-gw-02(config-if)# cofiguration terminal
% Unknown command: cofiguration terminal
msk-frank-gw-02(config-if)# exit
msk-frank-gw-02(config)# router ospf6
msk-frank-gw-02(config-ospf6)# ospf6 router-id 2.2.2.2
msk-frank-gw-02(config-ospf6)# exit
msk-frank-gw-02(config)# interface eth0
msk-frank-gw-02(config-if)# ipv6 ospf6 area 0
msk-frank-gw-02(config-if)# exit
msk-frank-gw-02(config)# exit
msk-frank-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-02#
```

Захват из Standard input [TabouFrank-sw-01 Ethernet1 to msk-frank-gw-01 eth0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

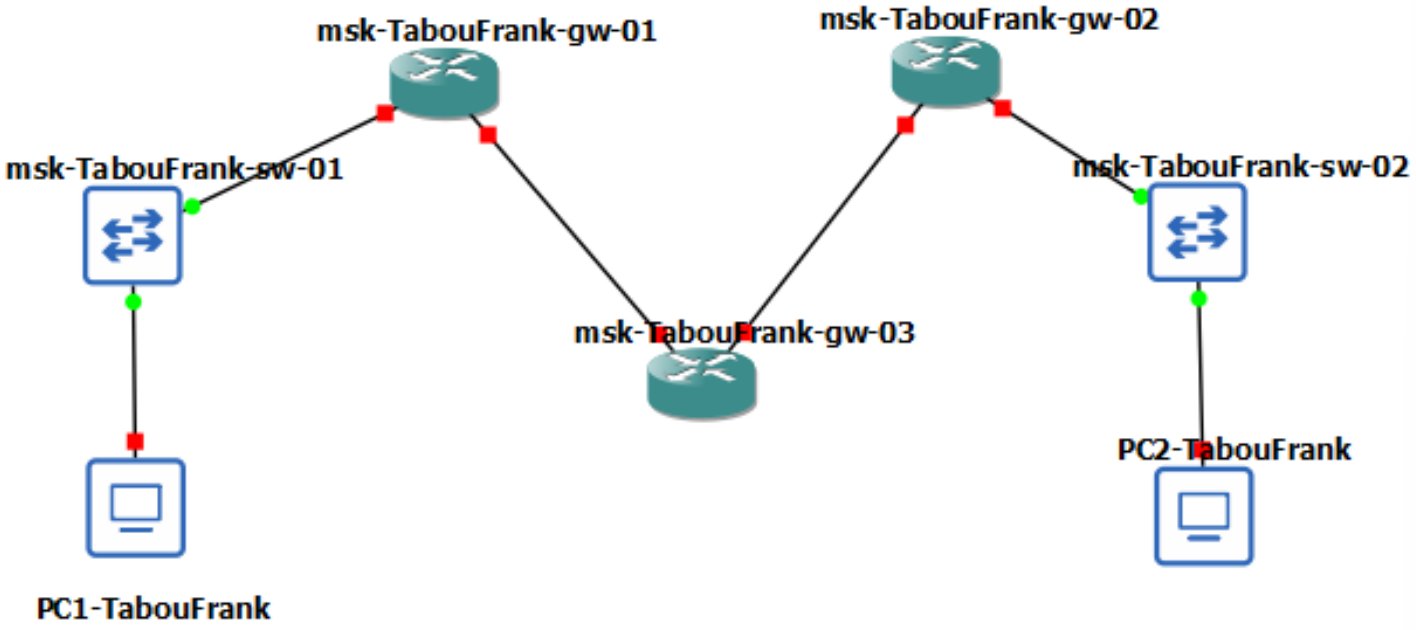
Примените фильтр отображения ... <Ctrl-/>

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 645 | 3198.808613 | 2001:10::a | 2001:11::a | UDP | 126 | 18480 → 18481 Len=64 |
| 646 | 3198.810608 | 2001:4::1 | 2001:10::a | ICMPv6 | 174 | Time Exceeded (Hop limit exceeded in transit) |
| 647 | 3198.811802 | 2001:10::a | 2001:11::a | UDP | 126 | 18480 → 18481 Len=64 |
| 648 | 3198.821050 | 2001:3::1 | 2001:10::a | ICMPv6 | 174 | Time Exceeded (Hop limit exceeded in transit) |
| 649 | 3198.821839 | 2001:10::a | 2001:11::a | UDP | 126 | 18480 → 18481 Len=64 |
| 650 | 3198.827434 | 2001:3::1 | 2001:10::a | ICMPv6 | 174 | Time Exceeded (Hop limit exceeded in transit) |
| 651 | 3198.828735 | 2001:10::a | 2001:11::a | UDP | 126 | 18480 → 18481 Len=64 |
| 652 | 3198.832468 | 2001:3::1 | 2001:10::a | ICMPv6 | 174 | Time Exceeded (Hop limit exceeded in transit) |
| 653 | 3198.833094 | 2001:10::a | 2001:11::a | UDP | 126 | 18480 → 18481 Len=64 |
| 654 | 3198.837054 | 2001:11::a | 2001:10::a | ICMPv6 | 174 | Destination Unreachable (Port unreachable)[Malformed Packet] |
| 655 | 3198.837282 | 2001:10::a | 2001:11::a | UDP | 126 | 18480 → 18481 Len=64 |
| 656 | 3198.841102 | 2001:11::a | 2001:10::a | ICMPv6 | 174 | Destination Unreachable (Port unreachable)[Malformed Packet] |
| 657 | 3198.841652 | 2001:10::a | 2001:11::a | UDP | 126 | 18480 → 18481 Len=64 |
| 658 | 3198.847822 | 2001:11::a | 2001:10::a | ICMPv6 | 174 | Destination Unreachable (Port unreachable)[Malformed Packet] |
| 659 | 3201.201688 | fe80::e2e:91ff:fe2f… | ff02::9 | RIPng | 106 | Command Response, Version 1 |
| 660 | 3203.498301 | fe80::e2e:91ff:fe2f… | ff02::5 | OSPF | 90 | Hello Packet |
| 661 | 3203.643707 | 10.0.10.1 | 224.0.0.5 | OSPF | 78 | Hello Packet |
| 662 | 3213.499565 | fe80::e2e:91ff:fe2f… | ff02::5 | OSPF | 90 | Hello Packet |
| 663 | 3213.644544 | 10.0.10.1 | 224.0.0.5 | OSPF | 78 | Hello Packet |
| 664 | 3220.436447 | 10.0.10.1 | 224.0.0.9 | RIPv2 | 126 | Response |
| 665 | 3223.500795 | fe80::e2e:91ff:fe2f… | ff02::5 | OSPF | 90 | Hello Packet |
| 666 | 3223.644952 | 10.0.10.1 | 224.0.0.5 | OSPF | 78 | Hello Packet |

```
Frame 1: 126 bytes on wire (1008 bits), 126 bytes captured (1008 bits) on interface -, id 0
    Section number: 1
  > Interface id: 0 (-)
    Encapsulation type: Ethernet (1)
    Arrival Time: Mar 13, 2026 20:10:55.995775000 RTZ 2 (зима)
    UTC Arrival Time: Mar 13, 2026 17:10:55.995775000 UTC
    Epoch Arrival Time: 1773421855.995775000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 0.000000000 seconds]
    [Time delta from previous displayed frame: 0.000000000 seconds]
    [Time since reference or first frame: 0.000000000 seconds]
    Frame Number: 1
    Frame Length: 126 bytes (1008 bits)
    Capture Length: 126 bytes (1008 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
```

```
0000  01 00 5e 00 00 09 0c 2e  91 2f 00 00 08 00 45 c0   ··^····· ·/····E·
0010  00 70 ba 10 40 00 01 11  ca a2 0a 00 0a 01 e0 00   ·p··@··· ********
0020  00 09 02 08 02 08 00 5c  d7 06 02 02 00 00 00 02   ·******\ ********
0030  00 00 0a 00 01 00 ff ff  ff 00 00 00 00 00 00 00   ******** ********
0040  00 01 00 02 00 00 0a 00  02 00 ff ff ff 00 00 00   ******** ********
0050  00 00 00 00 00 03 00 02  00 00 0a 00 03 00 ff ff   ******** ********
0060  ff 00 00 00 00 00 00 00  00 02 00 02 00 00 0a 00   ******** ********
0070  04 00 ff ff ff 00 00 00  00 00 00 00 00 01         ******** ******
```

```
msk-frank-gw-01# show ipv6 ospf6 router
% Unknown command: show ipv6 ospf6 router
msk-frank-gw-01# show ipv6 ospf6 route
*N IA 2001:1::/64                    ::                          eth1 00:03:42
*N IA 2001:2::/64                    fe80::e12:bfff:fe9d:1       eth2 00:03:42
*N IA 2001:3::/64                    fe80::e12:bfff:fe9d:1       eth2 00:03:42
*N IA 2001:4::/64                    ::                          eth2 00:03:42
*N IA 2001:10::/64                   ::                          eth0 00:03:42
*N IA 2001:11::/64                   fe80::e12:bfff:fe9d:1       eth2 00:03:42
msk-frank-gw-01# show ipv6 ospf6 route
*N IA 2001:1::/64                    ::                          eth1 00:08:02
*N IA 2001:2::/64                    fe80::e12:bfff:fe9d:1       eth2 00:08:02
*N IA 2001:3::/64                    fe80::e12:bfff:fe9d:1       eth2 00:08:02
*N IA 2001:4::/64                    ::                          eth2 00:08:02
*N IA 2001:10::/64                   ::                          eth0 00:08:02
*N IA 2001:11::/64                   fe80::e12:bfff:fe9d:1       eth2 00:08:02
msk-frank-gw-01#
```

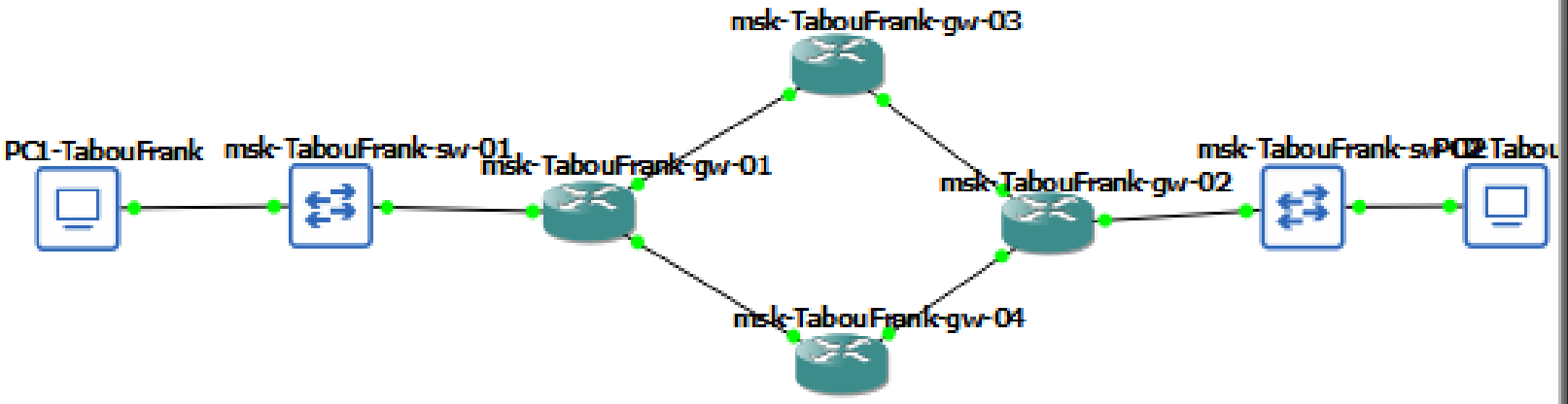

msk-TabouFrank-gw-01
msk-TabouFrank-gw-02
msk-TabouFrank-sw-01
msk-TabouFrank-sw-02
msk-TabouFrank-gw-03
PC2-TabouFrank
PC1-TabouFrank

PC2-TabouFrank - PuTTY

```
VPCS> ip 1002::a/64
PC1 : 1002::a/64

VPCS> save
Saving startup configuration to startup.vpc
.  done

VPCS> sahow ipv6
Bad command: "sahow ipv6". Use ? for help.

VPCS> show ipv6

NAME                : VPCS[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE        : 1002::a/64
DNS                 :
ROUTER LINK-LAYER   :
MAC                 : 00:50:79:66:68:01
LPORT               : 10020
RHOST:PORT          : 127.0.0.1:10021
MTU:                : 1500

VPCS>
```

msk-TabouFrank-gw-01 - PuTTY

```
You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name msk-TabouFrank-gw-0x
[edit]
vyos@vyos# compare
[edit system]
>host-name msk-TabouFrank-gw-0x
[edit]
vyos@vyos# commit
sa[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
eDone
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$ reboot
Are you sure you want to reboot this system? [y/N] y

vyos@vyos:~$
```


msk-TabouFrank-gw-03
PC1-TabouFrank
msk-TabouFrank-sw-01
msk-TabouFrank-gw-01
msk-TabouFrank-gw-02
msk-TabouFrank-gw-04

```
[   20.318095] systemd[1]: Detected architecture x86-64.
[   20.341849] systemd[1]: Set hostname to <vyos>.
[   21.077323] systemd[1]: Listening on udev Control Socket.
[   21.087302] systemd[1]: Listening on udev Kernel Socket.
[   21.097306] systemd[1]: Created slice system-serial\x2dgetty.slice.
[   21.114478] systemd[1]: Created slice User and Session Slice.
[   21.141350] systemd[1]: Started Dispatch Password Requests to Console Directo
ry Watch.
[   21.154598] systemd[1]: Listening on Journal Socket.
[   21.361369] bridge: filtering via arp/ip/ip6tables is no longer available by
default. Update your scripts to load br_netfilter if you need this.
[   21.384653] Bridge firewalling registered
[   21.405751] mpls_gso: MPLS GSO support

Welcome to VyOS - vyos ttyS0

vyos login:
```

msk-TabouFrank-gw-01 - PuTTY

```
[   21.405751] mpls_gso: MPLS GSO support

Welcome to VyOS - vyos ttyS0

vyos login: vyos
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$ configure
[edit]
vyos@vyos# show interfaces
 ethernet eth0 {
     address dhcp
     hw-id 0c:b5:af:f4:00:00
 }
 ethernet eth1 {
     hw-id 0c:b5:af:f4:00:01
 }
 ethernet eth2 {
     hw-id 0c:b5:af:f4:00:02
 }
 loopback lo {
 }
[edit]
vyos@vyos#
```

msk-TabouFrank-gw-01 - PuTTY

```
You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$ configure
[edit]
vyos@vyos# show interfaces
 ethernet eth0 {
     address dhcp
     hw-id 0c:b5:af:f4:00:00
 }
 ethernet eth1 {
     hw-id 0c:b5:af:f4:00:01
 }
 ethernet eth2 {
     hw-id 0c:b5:af:f4:00:02
 }
 loopback lo {
 }
[edit]
vyos@vyos# set interfaces ethernet eth0 address 2011:db8:1:1::1/64
[edit]
vyos@vyos# set interfaces ethernet eth1 address 2001:db8:1:1::1/64
[edit]
vyos@vyos# set interfaces ethernet eth0 address 2001:db8:1:1::1/64
[edit]
vyos@vyos# set interfaces ethernet eth1 address 2001:db8:1:2::1/64
[edit]
vyos@vyos# set protocols ospf area 0 network 10.10.1.0/24
[edit]
vyos@vyos#
```

# *Вывод*

В ходе выполнения лабораторной работы были изучены основные принципы маршрутизации в сетях IPv4 и IPv6, а также особенности настройки сетевого оборудования. Была построена модель сети, настроена адресация интерфейсов и проверена связь между узлами. Кроме того, были изучены принципы работы динамической маршрутизации и проведена проверка корректности передачи пакетов между сегментами сети. Полученные результаты показали, что правильная настройка маршрутизации обеспечивает эффективный обмен данными и стабильную работу сети.

*Спасибо за внимание !*